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UECM3473OE3b

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Review of preview

Started on	Friday, 24 July 2026, 11:37 AM
Completed on	Friday, 24 July 2026, 11:37 AM
Time taken	13 secs
Grade	0 out of a maximum of 10 (0%)

1 

Marks: 2

Assume an individual insured is selected at random from a population of insureds. The number of claims (N) experienced in a given year by each insured follows a distribution with the following probability density function (pdf):

$$f(n|\theta) = \theta^n e^{-\theta}/n!, \quad n = 0, 1, \dots,$$

where θ is randomly distributed across the population according to the following distribution:

$$f(\theta) = 6^2/\Gamma(2)\theta^1 e^{-6\theta}, \quad \theta > 0.$$

You have the following information on the number of claims made by an insured in the past 6 years:

0, 2, 3, 0, 2, 0

Find the probability that there is no claim next year. _____

Answer:



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Incorrect

Correct answer: 0.4866

Marks for this submission: 0/2.

2 

Marks: 2

Claim sizes are normally distributed with mean θ and variance 100,000. θ varies by risk, and is normally distributed with mean 1,500 and variance 1,010,000. For a certain risk, 12 claims averaging 2200 are observed. Determine the posterior probability that θ is less than 2267.0. _____

Answer:



[Make comment or override grade](#)

Incorrect

Correct answer: 0.7881

Marks for this submission: 0/2.

3 

Marks: 1

The number of claims per year on an insurance coverage has a binomial distribution with parameter $m = 3$ and Q . Q varies by insured and is distributed according to the following density function:

$$f(q) = cq(1 - q)^{11}, 0 \leq q \leq 1,$$

where c is a constant.

An insured submits 1 claims in 9 years. Calculate the posterior probability that for this insured, Q is less than 0.04. _____

Answer:



[Make comment or override grade](#)

Incorrect

Correct answer: 0.214465

Marks for this submission: 0/1.

4 

Marks: 1

We assume that the amount of an individual claim, Y , follows an exponential distribution function with probability density function

$$f(y|\delta) = 1/\delta e^{-y/\delta}, y, \delta > 0$$

The mean claim amount, δ , follows an inverse gamma distribution with density function

$$\pi(\delta) = 3^4 e^{-3/\delta} / (\Gamma(4) \delta^5), \delta > 0$$

Suppose 7 claims are observed with total aggregate claim amount of 34. Find $P(Y_8 > 1 | \sum Y_i = 34)$. _____

Answer:



[Make comment or override grade](#)

Incorrect

Correct answer: 0.7458
Marks for this submission: 0/1.

5

Marks: 1

Claim size for each insured has an exponential distribution with mean γ . γ varies according to an inverse gamma distribution with parameters $\alpha = 3$, $\theta = 1900$. An insured is selected at random. Claim experience for this insured is as follows:

Year 1: 1 claim for 650

Year 2: 2 claims for 570, 700

Determine the posterior estimate of average claim size for this insured. _____

Answer:

X

[Make comment or override grade](#)

Incorrect

Correct answer: 764

Marks for this submission: 0/1.

6

Marks: 1

Losses follow a distribution with density function

$$f(x) = \delta x^{\delta-1}, 0 \leq x \leq 1$$

δ varies by insured according to a gamma distribution with $\alpha = 4$, $\theta = 9$. A loss size of 0.55 is observed. Determine the posterior estimate of δ using zero-one loss function. _____

Answer:

X

[Make comment or override grade](#)

Incorrect

Correct answer: 5.6422

Marks for this submission: 0/1.

7

Marks: 1

The annual aggregate claim amounts (X) for the past 5 years for a risk is

269, 262, 268, 259, 242.

Suppose $X|\theta \sim N(\theta, v = 390)$ and $\theta \sim N(\mu = 250, a = 200)$, determine the lower bound of the 90% HPD credibility interval for θ . _____

Answer:

X

[Make comment or override grade](#)

Incorrect

Correct answer: 244.9
Marks for this submission: 0/1.

8 

Marks: 1

In a Bayesian analysis, a parameter θ has a continuous posterior with pdf
$$\pi(\theta) = ce^{-0.11\theta} \text{ for } 1 < \theta < 8; 0 \text{ otherwise,}$$

for an appropriate constant c . Find the upper bound of a 94% "HPD" credible set for θ . _____

Answer:



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Incorrect

Correct answer: 7.38844

Marks for this submission: 0/1.



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